Mohammed Sufiyan

Boston, MA (open to relocate) | LinkedIn | 857-263-1400
suff280602@gmail.com | GitHub | Google Scholar | Portfolio

EDUCATION

Boston University — **Boston, MA**
*MS in Applied Data Analytics (**GPA: 3.9**)* — *Sep 2025 - Dec 2026*

Presidency University — **Bengaluru, India**
B Tech in Computer Science and Engineering (Data Science) — *Sep 2020 - Jul 2024*

WORK EXPERIENCE

Power Tech — **Bengaluru, India**
Data Engineer — *Jun 2024 - May 2025*

- Cut processing time by **40%** with **25+ Python** cleaning and transformation workflows across **17 datasets**.
- Improved reporting performance by **30%** through **40+ SQL queries** and **5 Power BI dashboards** for daily operations.
- Processed **10GB+ daily** with AWS S3 and Lambda ETL and containerized **4 apps** with Docker, cutting deployment from **3 hours to under 30 minutes**.

Boston University — **Boston, MA**
Graduate Teaching Assistant | MET CS 688 — *Sep 2026 - Present*

- Grade technical assignments and explain coursework through Zoom sessions for **35 students**.

Boston University | GitHub | Live Demo — **Boston, MA**
Graduate Research Assistant | Urban Heat Island and CleanCampus — *Jan 2026 - Aug 2026*

- Designed a **29 table PostgreSQL** schema and Node.js REST APIs to validate and ingest campus sensor readings.
- Analyzed temperatures in MATLAB and deployed a Leaflet dashboard on DigitalOcean to display sensor maps, temperature trends, and missing-data alerts.
- Presented Urban Heat Island research at Boston University and **ACSP 2026** in Pittsburgh, Pennsylvania.

PROJECTS

Card Fraud Detection and Alerting Platform | GitHub — *Oct 2026*

- Scored **10,000+ simulated card transactions in a local test** with Python rules for amount, merchant category, country, channel, time, and device signals, returning a score and reason for each.
- Implemented a Redpanda streaming pipeline with PostgreSQL score storage and Redis alert caching, packaged with Docker Compose for API and dashboard access.

Pipeline Health Monitor | GitHub — *Sep 2026*

- Validated **100,000 synthetic orders in a local test** with **Apache Spark** and orchestrated quality checks using **Apache Airflow**.
- Built **dbt** models and tests in **Snowflake**, tracked pipeline runs and rejected records in MySQL and Streamlit, packaged with Docker Compose.

Formula 1 Pit Stop Analysis and Prediction | GitHub — *Feb 2026*

- Modeled **54 SQL Server tables** and a star schema, using Python ETL to clean **10,452 records with 48 attributes** and add indexes and season partitioning.
- Compared Logistic Regression, SVM, and **Random Forest** in scikit-learn. Random Forest led slow pit stop prediction with **93.8% test accuracy** and **0.875 F1**.

Multi Agent Web Research Assistant | GitHub — *Jan 2026*

- Developed a **4 agent LangGraph workflow** with Llama 3.2 through Ollama, using two researchers, an evaluator, and a writer to search, deduplicate, rank evidence, and return cited answers.

Research Evidence Explorer | GitHub — *Dec 2025*

- Extracted **21 method, dataset, and metric entities** from **8 sample paper records** in a local Python demo and implemented PostgreSQL storage and FastAPI paper search.

RESEARCH PAPERS

Pix2Pix GAN Lung Segmentation for Chest Radiographs | **Published in IEEE** | IEEE — *Sep 2024 - Apr 2025*

- Trained a Pix2Pix U-Net with TensorFlow and OpenCV, achieving **98.25% accuracy**, **98.05% Dice** on Montgomery, and **95.87% accuracy** on Shenzhen.

LLMs in Reinforcement Learning Comparative Analysis | **Accepted in IEEE** — *Nov 2025 - Present*

- Integrated Llama 2 7B and Sentence-BERT with PPO, raising mean DoorKey reward from **0.38 to 0.53** across **8 runs** per method using policy guidance and reward shaping.

SKILLS

Data Engineering & Databases: Python, Pandas, SQL, Apache Airflow, Apache Spark, Snowflake, dbt, PostgreSQL, SQL Server, MySQL, Redis, Redpanda, ETL, Data Validation, Data Modeling
Cloud & Tools: AWS S3, AWS Lambda, Docker, Docker Compose, DigitalOcean, PM2, nginx, Power BI, Git
AI / ML & Backend: scikit-learn, TensorFlow, OpenCV, LangGraph, Ollama, Llama, Sentence-BERT, PPO, MATLAB, Node.js, FastAPI, Streamlit, REST APIs